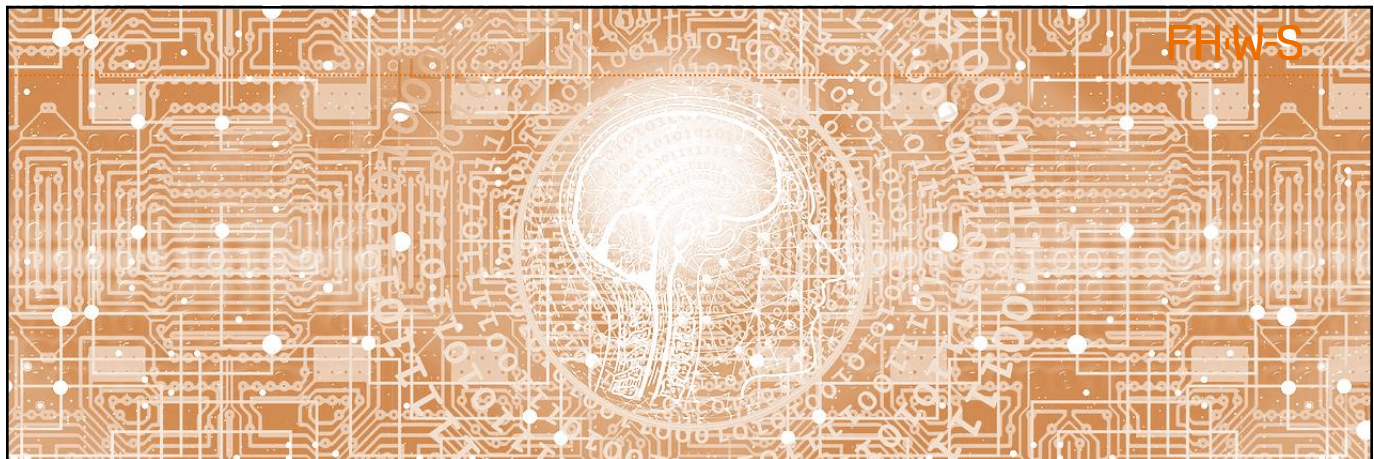




Software Engineering and Cyber Security

Prof. Dr. Marian Daun

FHWS University of Applied Sciences
Würzburg-Schweinfurt

The slide has a background image of a desk with a smartphone, a pen, and an open notebook. The text 'Agenda' is in a large orange font at the top left. Below it, a list of topics is enclosed in a white box with an orange border. The FHWS logo is at the bottom left, and a small orange number '2' is at the bottom right.

Agenda

- Software Engineering Fundamentals
- Requirements Engineering
- Architecture Design
- Modelling
- Development and Implementation
- Software Quality Assurance

FHWS University of Applied Sciences
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2

Modelling

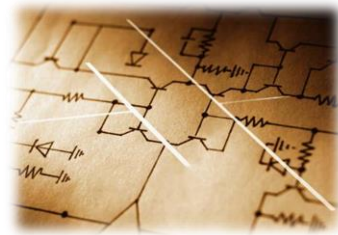
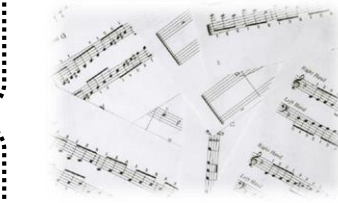
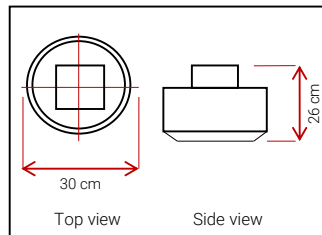
Part A: Conceptual Modelling

The Term Model

What is a Model?

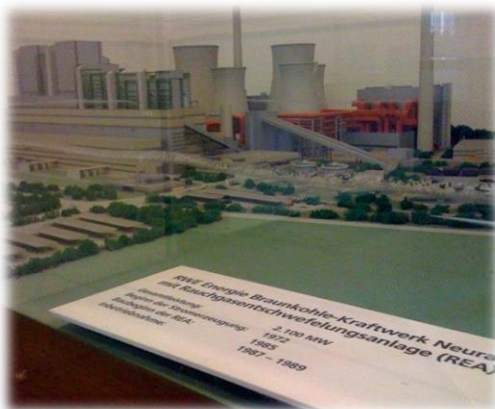
D A model is an abstract representation of the universe of discourse created for a specific purpose (use). Based on [Stachowiak 1973]

D The universe of discourse comprises any part or aspect of the existing or conceived reality under consideration. [Falkenberg et al. 1998]

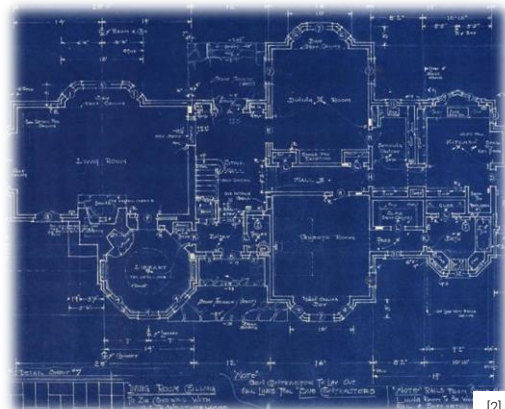


Examples

Physical Model



Conceptual Model





Group Discussion

Why do we need Models?

Models are Abstractions for a Specific Purpose

Universe of discourse:
A village



Task:
How to reach my home?

*if you have reached the huge
intersection, turn right*



then drive through

*If you have passed the village
name sign, turn into the next
street on the right hand side*



*Our house is the second one
on the left hand side*

Street Map of England

- Streets are not to scale
- Focus on main streets
- Only bigger cities
- Maritime borders
- Ferry lines



Street Map of London

- Different street types
- City quarters
- Housing
- Railway lines



Sightseeing Map of London

- Sights symbols
- Street names
- Streets straightened
- Parks and water highlighted



© Inside Guide to London [5]

Underground Map of London

- Metro lines (coloured)
- Metro stations
- Change between lines
- ...

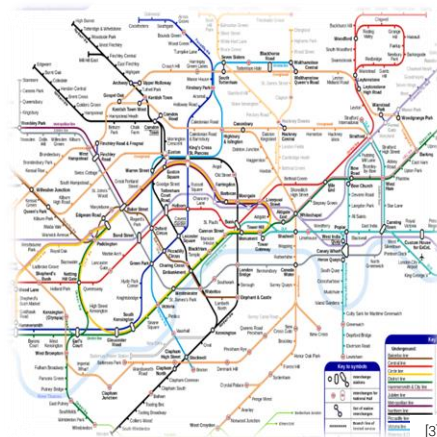


Different Abstractions of the Same Reality

Street map



Underground map



Characteristics of Models

- A model is always defined for a given purpose
 - The content of a model depends on its purpose
 - Different conceptual models support different purposes
 - Different models can overlap, but do not have to.
- Models differ in their level of abstraction and detail (e.g., street maps and underground maps)



Group Discussion

Why Use Models in Software Engineering?

Properties of Conceptual Models

Descriptive vs. Prescriptive Models



Descriptive models

Document information about the current state of the universe of discourse



Prescriptive models

Represent an intended, future state of the universe of discourse.

Two Central Properties of Models



Reduction



Extension

Reduction Property

- Reduction

Abstraction mechanisms are used to filter out irrelevant aspects; only relevant aspects of the world are represented in the model.

- Abstraction mechanism

- Selection Particular aspects of the real world are represented in the model (selected), while other aspects are ignored.
- Aggregation Different aspects are combined and represented in the model in a condensed form.
- Classification / generalization Commonalities in the real world are generalized and objects are classified; differences might at the same time be ignored.

Reduction Property

Selection

Only roads and settlements.

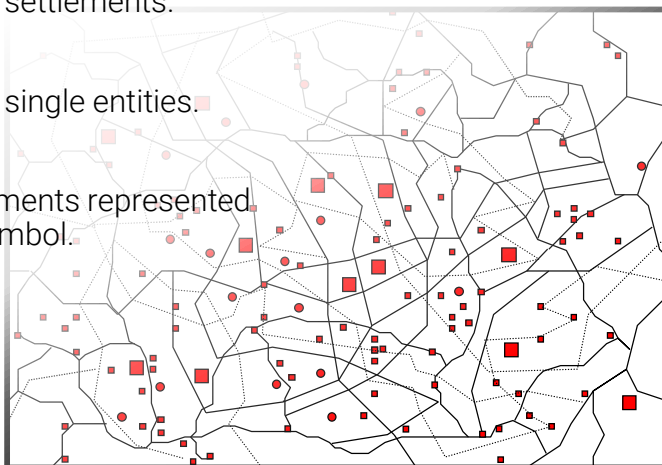
Aggregation

Settlements as single entities.

Classification

Different settlements represented by the same symbol.

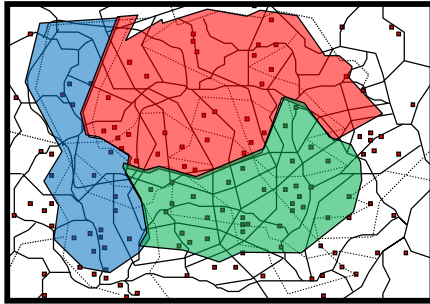
■	population < 10,000
●	population > 10,000
■	population > 100,000



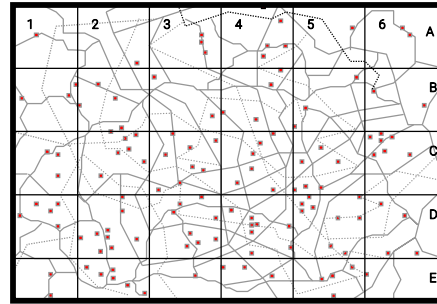
Extension Property

Additional information not observable in the universe of discourse is added (e.g., to improve the use of the model).

- Improve the comprehension by shapes.



- Improve the usability through quadrants.



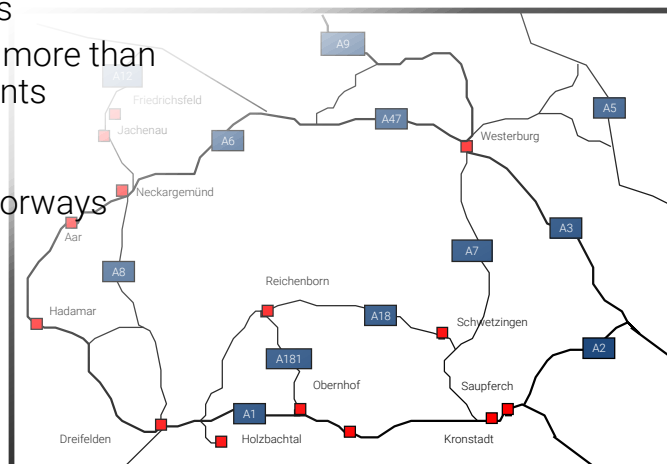
Reduction and Extension

Selection

- Only main roads
- Only cities with more than 50,000 inhabitants

Extension

- Number of motorways



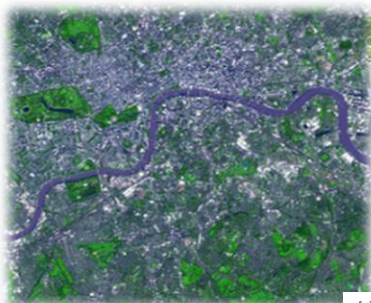


Group Discussion

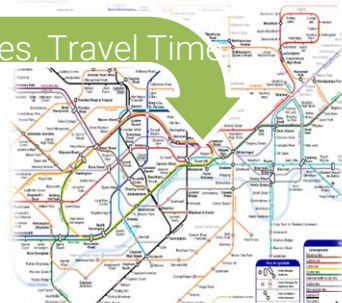
What Reductions and Extensions Are Used in Software Engineering

Modelling Language

- Models are created using a modelling language
- The modelling language is designed for a specific purpose
- Information is documented using the modelling constructs of the language



Stations, Lines, Travel Time



Reality
(universe of discourse)

Purpose-driven abstraction
(representation)

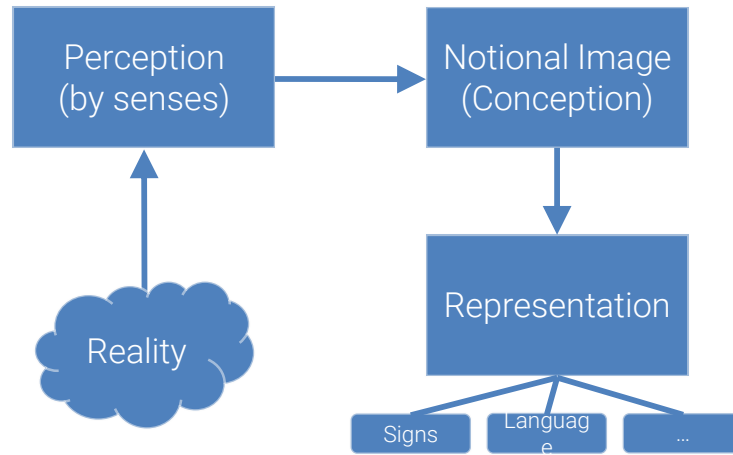


Group Discussion

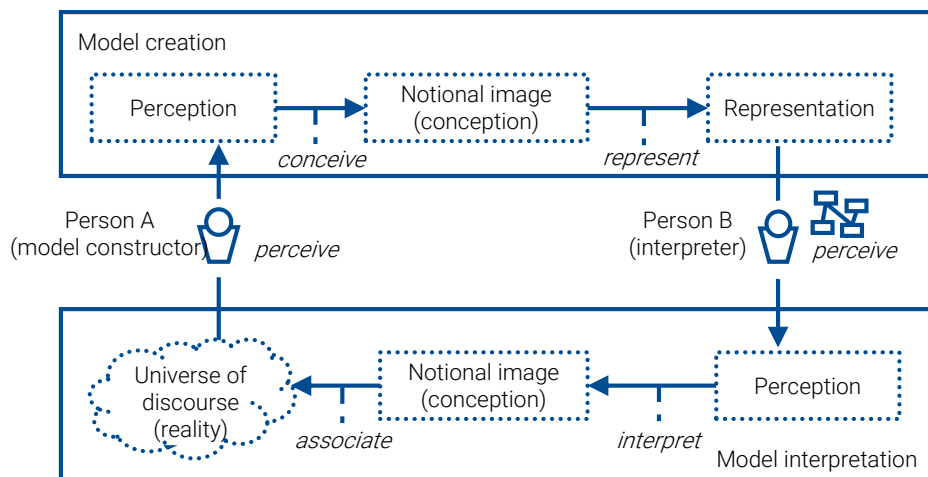
What Modelling Languages does a Software Engineering need to know?

Fundamentals of Conceptual Modelling

From Reality to Representation

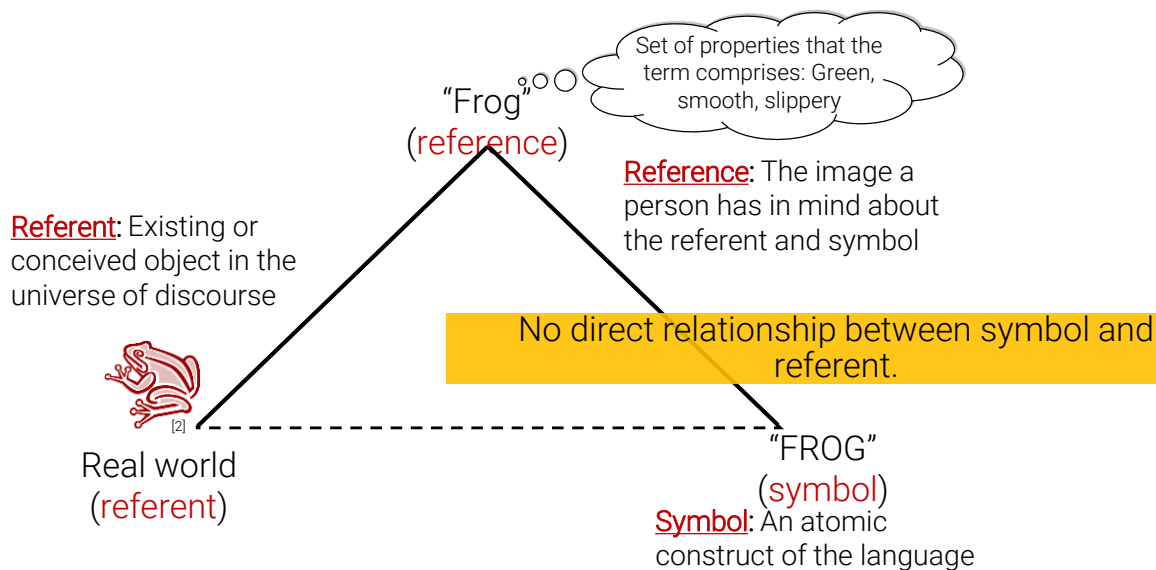


Model Construction and Interpretation



Persons A and B need to speak the same language.
Only then, the interpreter can associate the elements represented in the
language with the same aspects in reality.

The Semiotic Triangle



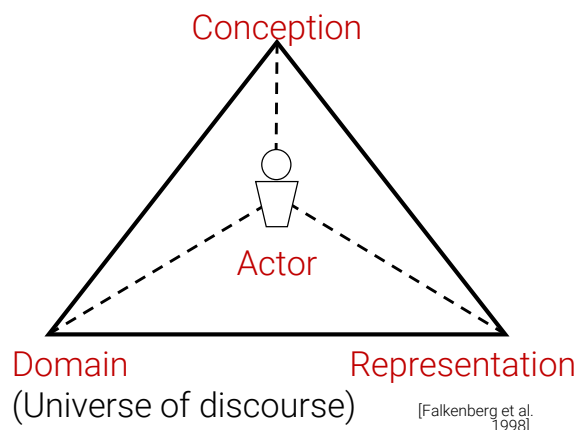
The Semiotic Tetrahedron

Domain:
Universe of discourse.

Conception:
Image of the real-world object in the mind of the interpreting person.

Representation:
Result of a human actor describing his/her conceptions.

Actor:
Model constructor or interpreter (of models).



Syntax, Semantics and Pragmatics

Syntax:

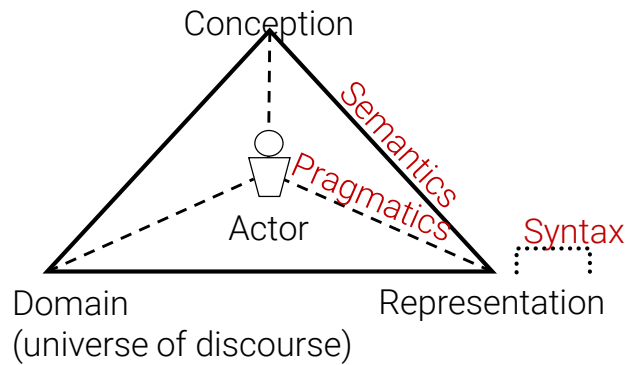
Atomic language constructs and their valid combinations.

Semantics:

Meaning of the language constructs and meaning of combined constructs is defined with composition principle.

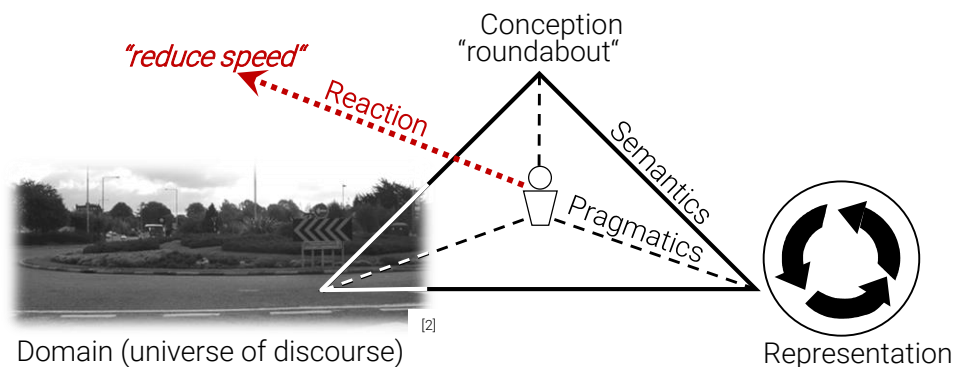
Pragmatics:

Deal with effects of the symbol on the behaviour of the interpreting actor.



[Falkenberg et al.
1998]

Syntax, Semantics and Pragmatics

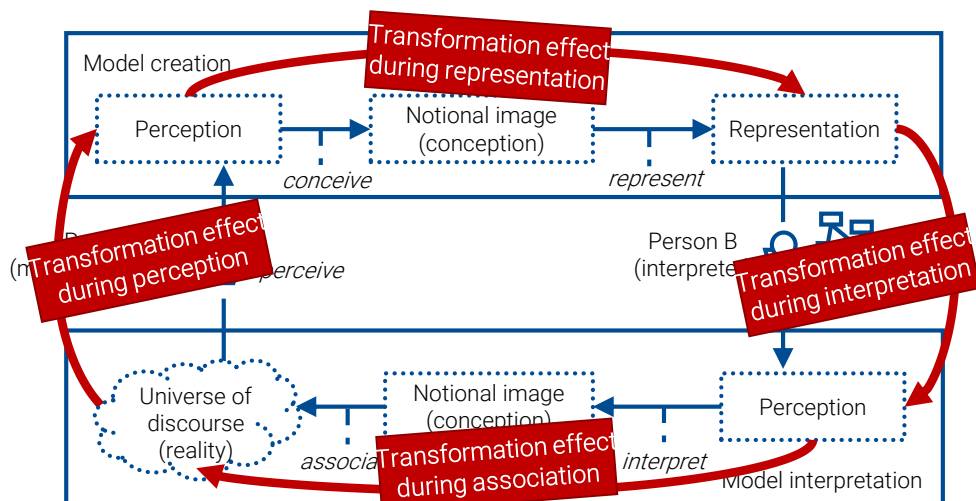




Group Discussion

What can go wrong?

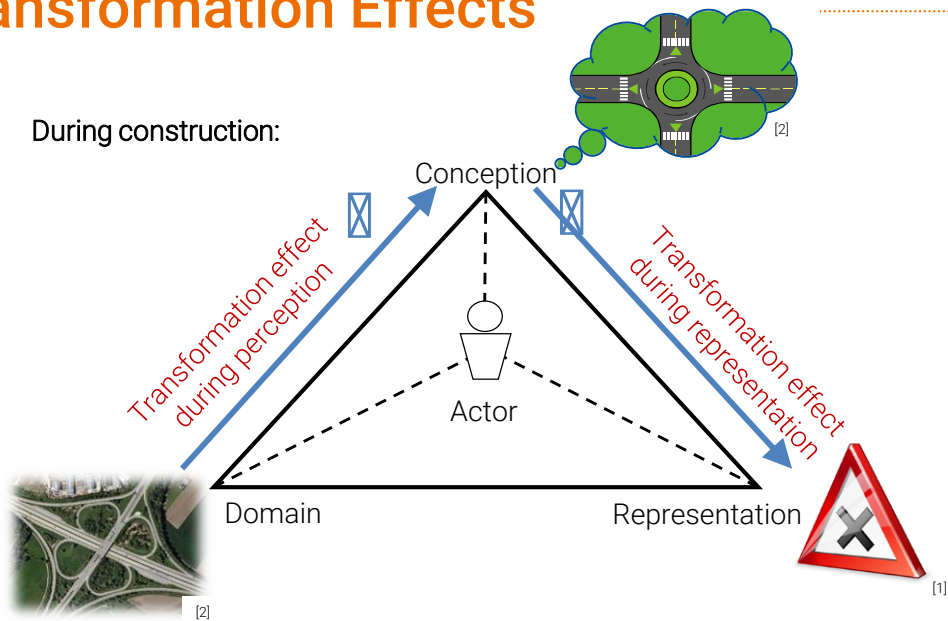
Four potential Transformation Effects



Persons A and B need to speak the same language.
Only then, the interpreter can associate the elements represented in the language with the same aspects in reality

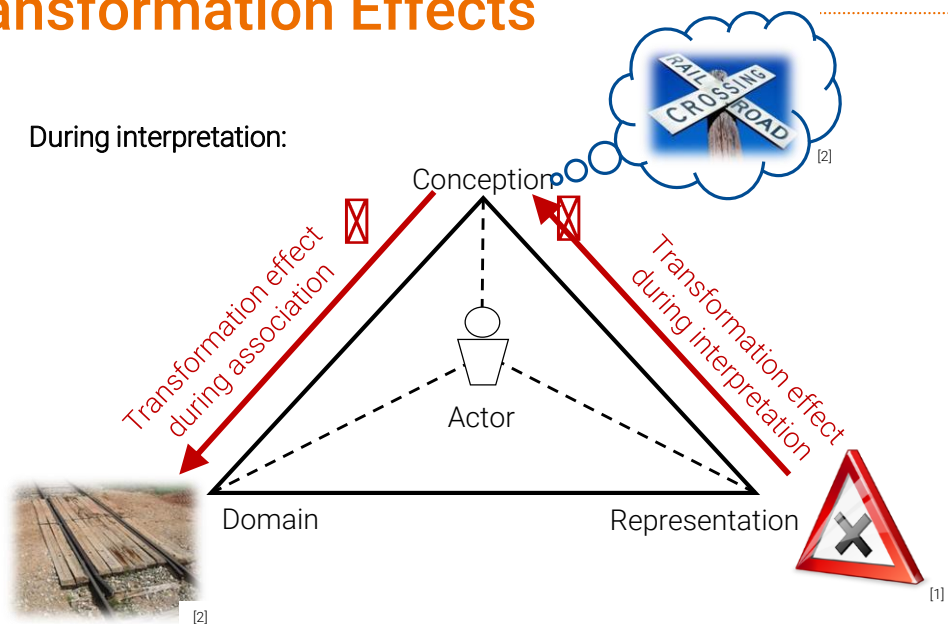
Transformation Effects

During construction:



Transformation Effects

During interpretation:



Quality of Conceptual Models

- Syntactic quality:
Adherence to syntactic rules
- Semantic quality:
Validity and completeness of the model
- Pragmatic quality:
Suitability for intended use

Deficiencies in one quality aspect often affect the other quality aspects of a conceptual model!

Key Benefits of Conceptual Models

- Focus on **specific purpose**
- Easy to **comprehend** and **memorize**
- **Reduction of complexity**
- Foster communication by **purpose-based abstractions**
- Support of **problem understanding and solving**
- Support for **automation** by partial formalization, e.g., for quality assurance activities

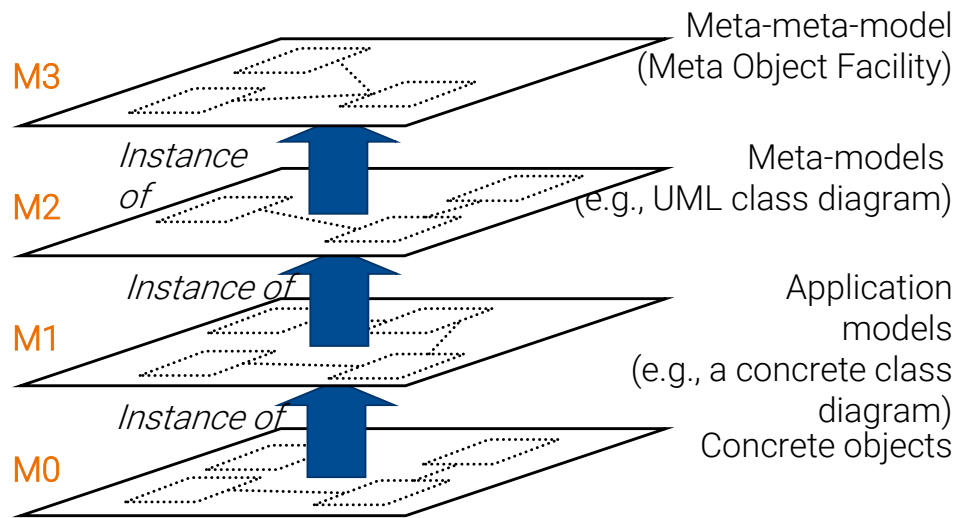


Group Discussion

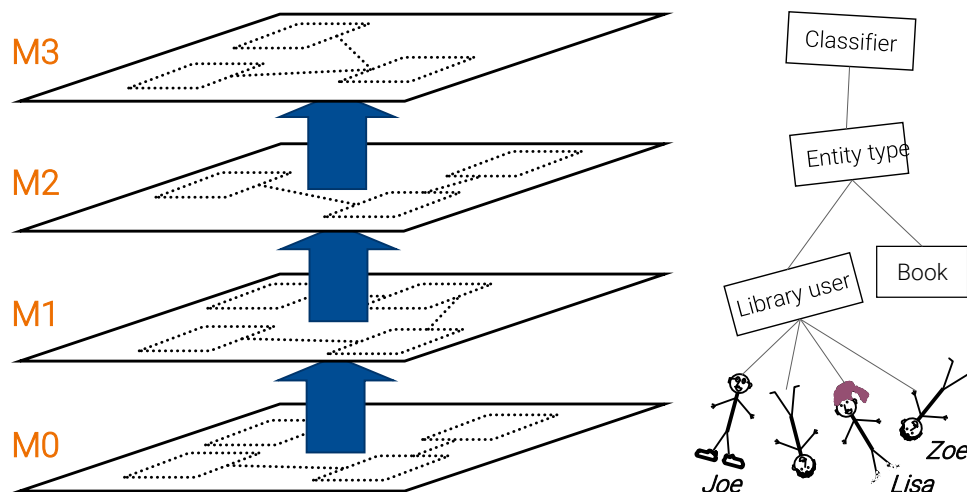
How Important is Formalization and Automation?

Meta Object Facility (MOF)

Four Modelling Layers



Example



Literature

- [Falkenberg et al. 1998] E. D. Falkenberg, W. Hesse, P. Lindgreen, B. E. Nilsson, J. L. Han Oei, C. Rolland, R. K. Stamper, F. J. M. Van Assche, A. A. Verrijn-Stuart, K. Voss: A Framework of Information System Concepts – The FRISCO Report. IFIP Report, 1998.
- [Pohl 2010] K. Pohl: Requirements Engineering - Fundamentals, Principles and Techniques, 1st ed., Springer, 2010.
- [Seidewitz 2003] E. Seidewitz: What Models Mean. IEEE Software, Vol. 20, No. 5, IEEE Press, Los Alamitos, 2003, pp. 26-32.
- [Stachowiak 1973] H. Stachowiak: Allgemeine Modelltheorie. Springer, Wien, 1973.
- [Ogden and Richards 1923] C. K. Ogden, I. A. Richards: The Meaning of Meaning – A Study of the Influence of Language upon Thought and of the Science of Symbolism. Routledge & Kegan, London, 1923 (new edition available: Harcourt Brace Jovanovich, 1989).

Questions for Self-Assessment

- What is the meaning of the term “model”?
- What are the typical properties of conceptual models?
- What is the role of abstraction in conceptual modelling?
- Which reduction principles do you know?
- Which extension principles do you know? What does it mean to “extend” a model? When does an extension makes sense?
- What is defined by the Semiotic Triangle and the Semiotic Tetrahedron?
- Which transformation effects can occur during model construction?
- Which transformation effects can occur during model interpretation?
- How can the quality of conceptual models be assessed?
- How are modelling languages defined?

Thanks

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The Ruhr Institute for Software Technology
<https://www.paluno.uni-due.de>

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